MATH 10A — STRUCTURE AND CONCEPTS IN MATH I

§5.1

Introduction to Integers

Opposites, the number line, charged chips, and absolute value

Today's Goal

After completing this section, you should be able to:

- Explain what the opposite of a number is, and why the opposite of a negative integer is a positive integer
- Identify the integers, and distinguish the positive integers, the negative integers, and the non-negative integers
- Represent an integer on the number line and with the charged chip model
- Interpret an absolute value as a distance on the number line, including reading $|n - a|$ as the distance between n and a

Learning Target 13

LT.13 Arithmetic with Integers

I can apply multiple strategies (including the standard algorithms) for addition, subtraction, multiplication, and division of integers using appropriate representations.

Adding to Get Zero

Which whole number can you add to 5 to get 0?

None of them. Adding any whole number to 5 gives a number that is at least 5.

To get 0, we need a new kind of number.

Opposite of a Whole Number

Definition 5.1.1

Given a whole number n, the **opposite of** n is a number denoted by ^{-}n such that $n + (^{-}n) = 0$.

Notes on Opposites

Note 5.1.2

1. In general, you will often see "$-n$" written when what is meant is "^{-}n". In these course notes, we will use "^{-}n" to avoid confusion with subtraction. However, in MyOpenMath, you will just write "$-n$" instead of "^{-}n".
2. The opposite of 0 is just 0 because $0 + 0 = 0$.
3. The opposite of any nonzero whole number n is no longer a whole number. This is because adding any whole number to n will result in a number that is at least n, which cannot be 0 because n is not 0.

Integers

Definition 5.1.3

The set of **integers** is the set of numbers made up of whole numbers and their opposites.

$$\ldots, {}^{-}4, {}^{-}3, {}^{-}2, {}^{-}1, \underbrace{0, 1, 2, 3, 4, \ldots}_{\text{Whole Numbers}}$$

Positive and Negative Integers

Note 5.1.5

1. The numbers $1, 2, 3, 4, \ldots$ are known as the **natural numbers** as well as the **positive integers**.
2. The numbers $^-1, ^-2, ^-3, ^-4, \ldots$ are known as the **negative integers**.
3. Zero is neither positive nor negative.

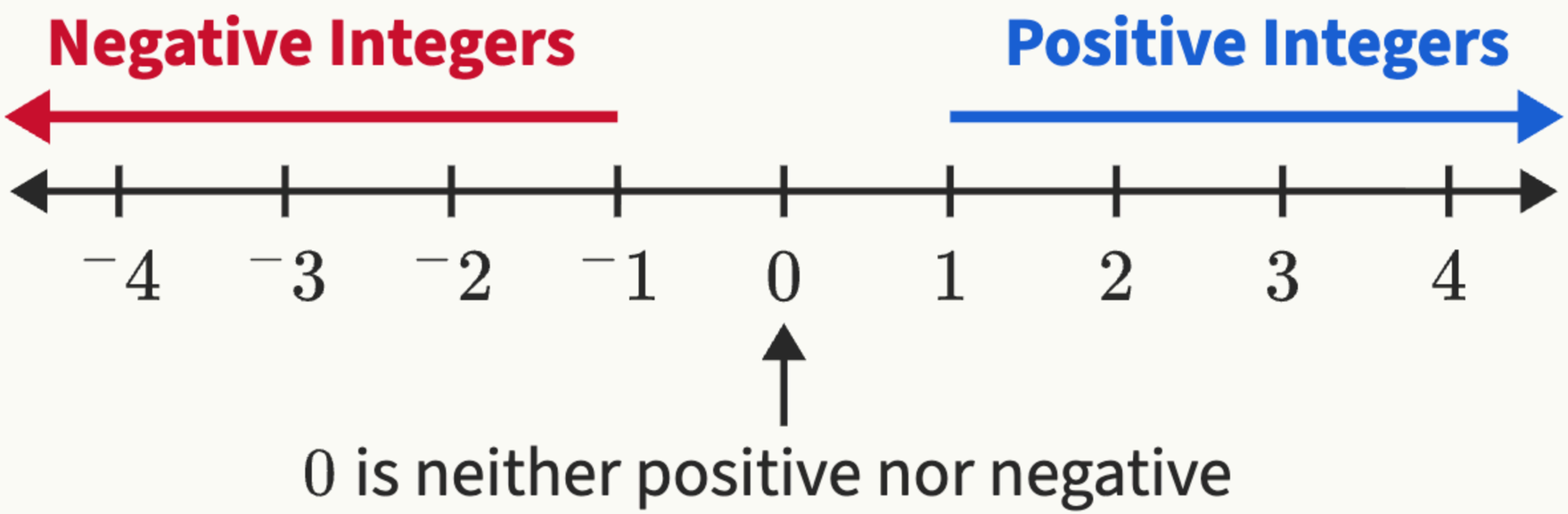
Non-negative Integers

Warning 5.1.6

Sometimes, we may refer to the set of **non-negative integers**. This is not the same as the **positive integers**. "Non-negative" means "not negative", so this would include 0 because it is an integer and it is not negative. In other words, the non-negative integers are $0, 1, 2, 3, 4, \ldots$.

The Number Line Model for Integers

The number line model we used previously for whole numbers can be extended to represent the integers as shown below.



Opposite of an Integer

Definition 5.1.8

Given an integer n, the **opposite of** n is a number denoted by ^{-}n such that $n + (^{-}n) = 0$.

Is ^{-}n Always Negative?

What is $^{-}(^{-}6)$?

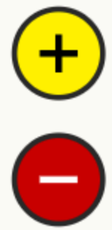
We know that $6 + (^{-}6) = 0$. On the one hand, this tells us that the opposite of 6 is $^{-}6$.

On the other hand, it tells us that the opposite of $^{-}6$ is 6. In other words, $^{-}(^{-}6) = 6$.

So ^{-}n is not always a negative number.

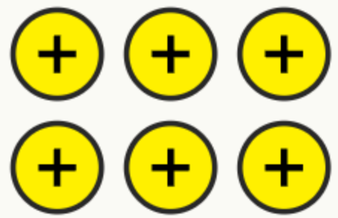
The Charged Chip Model for Integers

Another useful model for working with integers is the charged chip model. In this model, we use two different colored chips to represent positive and negative integers. In our case, we will use yellow chips (Y) for positive numbers and red chips (R) for negative numbers.

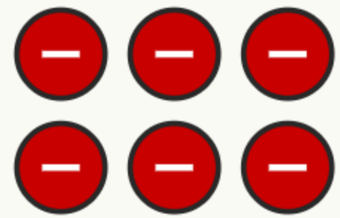


Yellow $= 1$

Red $= {}^{-}1$



6 represented using charged chips

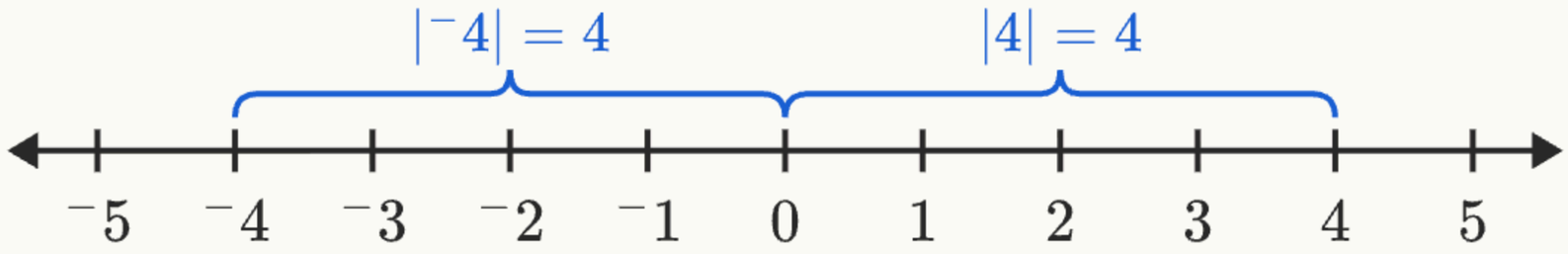


${}^{-}6$ represented using charged chips

Absolute Value

Definition 5.1.11

The **absolute value** of an integer n is the distance between n and 0 on the number line and is denoted by $|n|$.



Absolute Value and Zero

Warning 5.1.12

One common misunderstanding when interpreting absolute values is to think that the absolute value of n "makes n positive". While this is true if n is negative, it is not true when $n = 0$ because $|0| = 0$, which is not positive.

Absolute Value as a Distance

If we think of $|n|$ as $|n - 0|$, which are equivalent since $n = n - 0$, we can see where the 0 comes from in Definition 5.1.11.

This is a useful observation, since we could change 0 to another number and modify the meaning accordingly.

In general, we have $|n - a|$ representing the distance between n and a on the number line.

Interpreting Absolute Values

Interpret each of the following expressions using the definition of absolute values.

a. $|n-8|$

$|n-8|$ represents the distance between n and 8 on the number line.

b. $|8-n|$

$|8-n|$ represents the distance between 8 and n on the number line.

The order in which we list the numbers does not affect the distance between them. This tells us that $|n-8| = |8-n|$ and in general, $|n-a| = |a-n|$.

Same Distance Puzzle

Using the digits 1 to 9 at most one time each, fill in the boxes to make the following statement true.

$$\boxed{A} \text{ and } \boxed{B} \text{ are } \boxed{C} \text{ units away from } \boxed{D}.$$

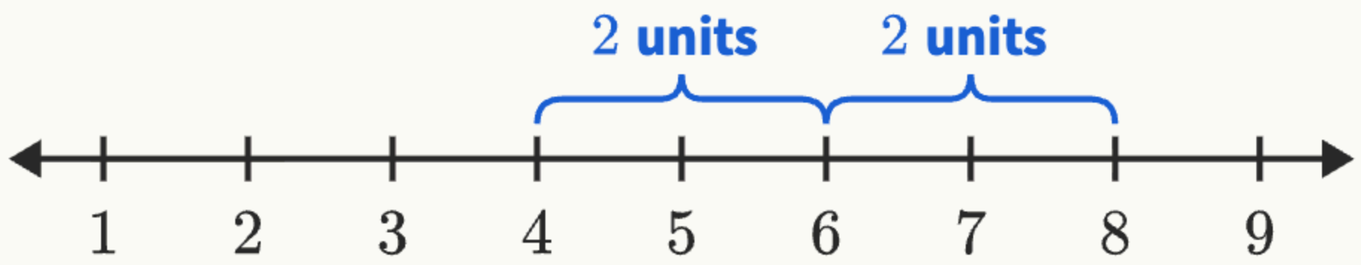
This statement means two things:

1. The distance between A and D is the same as the distance between B and D.
2. The distances mentioned above are both equal to C.

Same Distance Puzzle, Continued

One idea is to choose D to be somewhere in the middle first. Then choose C in a way that produces A and B so that they are all different digits.

For example, we could start with $D = 6$. Then we could choose $C = 2$. This means we would move 2 units to the left and 2 units to the right of 6 putting us at $A = 4$ and $B = 8$.



In other words, 4 and 8 are 2 units away from 6.

Find More Solutions

Find as many other solutions to Example 2 as you can.

There are 13 in all. Switching A and B gives the same statement.

A and B	C	D
1 and 5	2	3
2 and 4	1	3
1 and 7	3	4
3 and 5	1	4
1 and 9	4	5
2 and 8	3	5
3 and 7	2	5

A and B	C	D
4 and 6	1	5
4 and 8	2	6
5 and 7	1	6
5 and 9	2	7
6 and 8	1	7
7 and 9	1	8